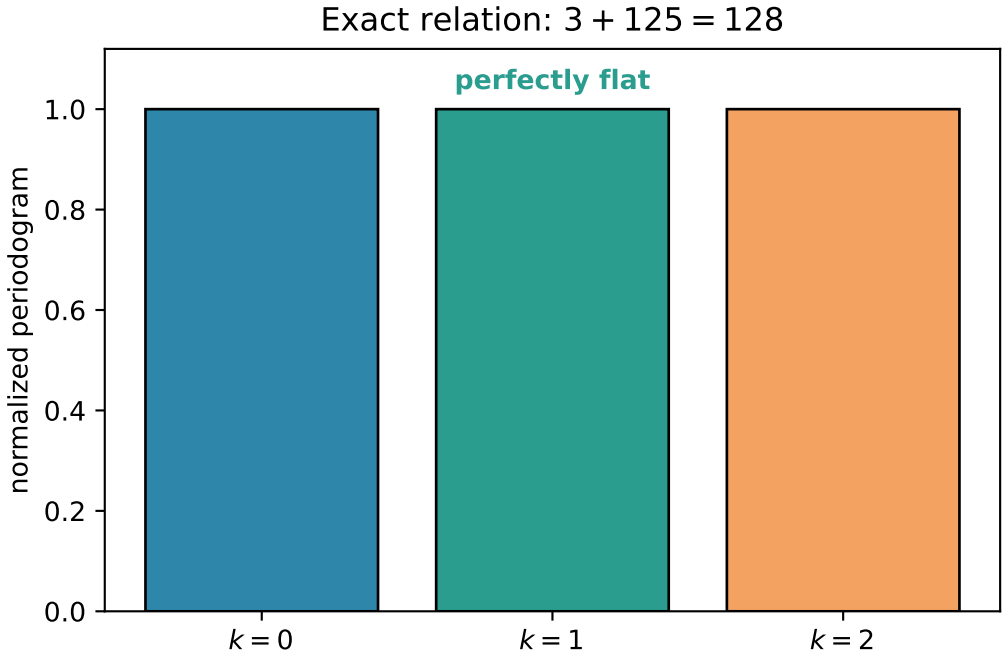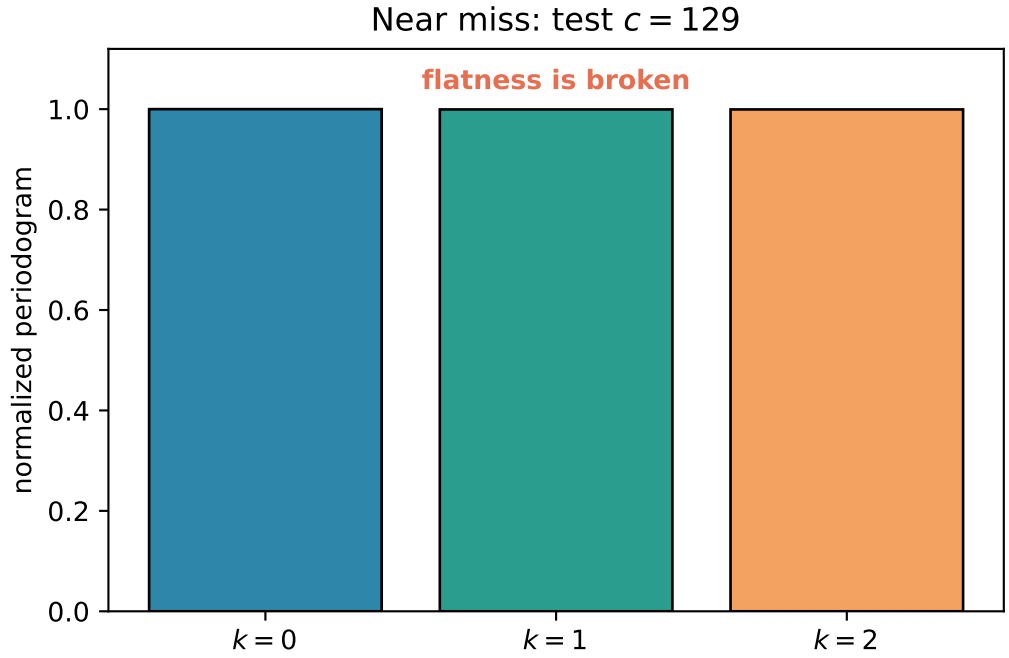The additive equation is a three-frequency flatness condition